

PROBLEM 1 (PAC)

GIVEN INPUTS	REQUIRED RESULTS
Student ID Student Name Subject 1 Marks Subject 2 Marks Subject 3 Marks	Total Obtained Marks Percentage
PROCESSING REQUIRED $total_obtained_marks = marks1 + marks2 + marks3$ $percentage = total_obtained_marks * 100 / 300$	SOLUTIONS 1. Use the Given Formula to find Percentage 2. Use total / 3 instead as simplified form

PROBLEM 2 (PAC)

GIVEN INPUTS	REQUIRED RESULTS
length width	Area of Rectangle
PROCESSING REQUIRED $area = length * width$	SOLUTION ALTERNATIVES 1. find area by multiplying length & width 2. check values for < 0 before finding area using length & width

PROBLEM 3 (PAC)

GIVEN INPUTS	REQUIRED RESULTS
temperature in Celsius	temperature in Fahrenheit
PROCESSING REQUIRED $temp_fahrenheit = (temp_celsius * 9 / 5) + 32$	SOLUTIONS 1. Use the standard given formula $(C * 9 / 5) + 32$ 2. Use the Alternate formula $(C * 1.8) + 32$ 3. Check for temperature being less than Absolute Zero

PROBLEM 4 (IPO)

INPUTS	PROCESSING	MODULE REFERENCE	OUTPUT
Monday Steps Tuesday Steps Wednesday Steps Thursday Steps Friday Steps	1. Enter Monday Steps 2. Enter Tuesday Steps 3. Enter Wednesday Steps 4. Enter Thursday Steps 5. Enter Friday Steps 6. Calculate Average 7. Output Average 8. END	Read Read Read Read Read Calc Print StepCountControl	Average Daily Steps

ALGORITHMS (P: 1-3)

PROBLEM 1

1. Ask the user to enter *subject 1 marks*
2. Ask the user to enter *subject 2 marks*
3. Ask the user to enter *subject 3 marks*
4. Set **total_obtained_marks** to (subject 1 marks + subject 2 marks + subject 3 marks)
5. Set **percentage** to (total_obtained_marks * 100 / 300)
6. Display **total_obtained_marks** for the user
7. Display **percentage** for the user

PROBLEM 2

1. Ask the user for *length*
2. Ask the user for *width*
3. Set *area* to (*length * width*)
4. Display *area* to the user

PROBLEM 3

1. Ask the user for *temperature in Celsius*
2. Set *temperature in Fahrenheit* to $[(\text{temperature in Celsius} * 9 / 5) + 32]$
3. Display *temperature in Fahrenheit* to the user